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Description - This material covers the replacement of the GRAM Analog Service Module Replacement, found only on the Signa Horizon HiSpeed and EchoSpeed systems.

1- INTRODUCTION

The GRAM Analog Service Module (GASM) Board (part no. 2103294) is a field-replaceable unit (FRU).

2- TOOLS REQUIRED

A DVM with alligator clips is required for the Lock Out Tag Out procedure.

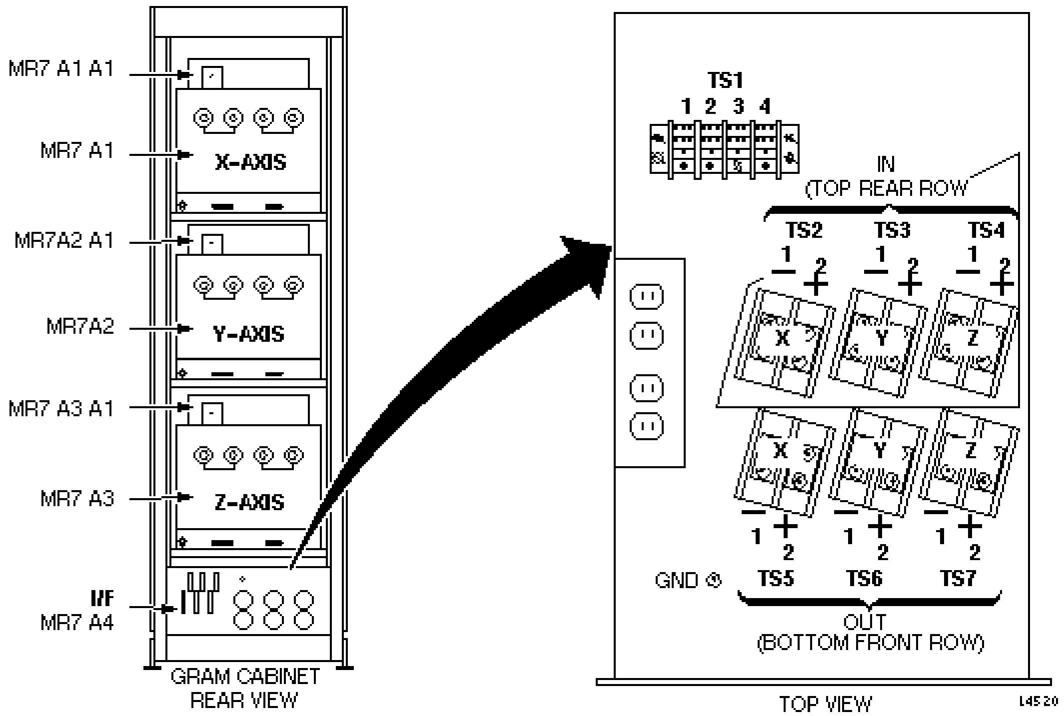
3- PRELIMINARY SET-UP

DANGER!!

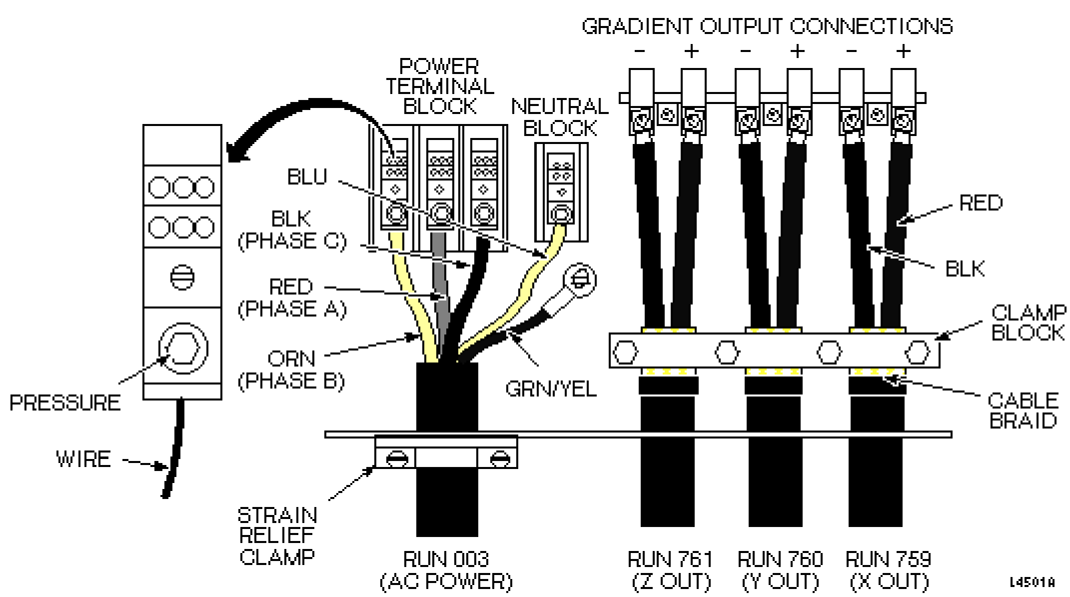
FATAL ELECTRIC SHOCK HAZARD!! THE GRAM AND GRADIENT AMPLIFIERS ACT AS CONSTANT LOAD SOURCES, AND WILL SEND MAXIMUM CURRENT TO ANY LOAD (INCLUDING YOU!). TO PREVENT FATAL ELECTRIC SHOCK, ENSURE THAT POWER IS OFF TO BOTH CABINETS BEFORE CONTINUING WITH THIS PROCEDURE.

1. Perform lock out / tag out by securing the PDU Circuit Breaker for the GRAM Cabinet and the 8645 Gradient Amplifier Cabinet with the lock out /tag out devices. (Refer to CD-ROM *Dir. 2187583-3 [or -2], MR Release Signa 5x/8x Service Methods, Renewal Parts and Service Tools,* Procedure for Safety, Section 6, OSHA LOCKOUT/TAGOUT REQUIREMENTS.

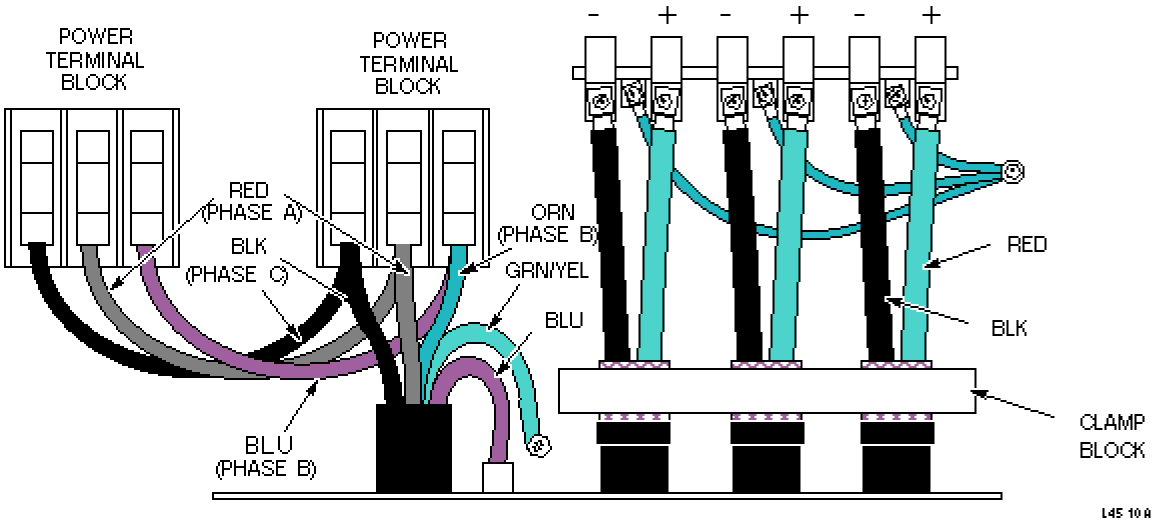
2. Verify that all energy has been dissipated by measuring incoming power to the GRAM Cabinet at TS1 (see Illustration L4520B). Verify that all energy has been dissipated for the 8645 Gradient Amplifier Cabinet by measuring power at TS1. Also see Illustration L4501A for Signa Horizon HiSpeed system, or Illustration L4510A for Signa Horizon or Horizon EchoSpeed systems.



GRAM CABINET, REAR VIEW – BOTTOM PANEL AND TS1
ILLUSTRATION L4520B



8645 CABINET POWER AND OUTPUT CABLE CONNECTIONS
ILLUSTRATION L4501A



INCOMING POWER TO A DOUBLE-BAY 8645 GRADIENT CABINET
ILLUSTRATION L4510A

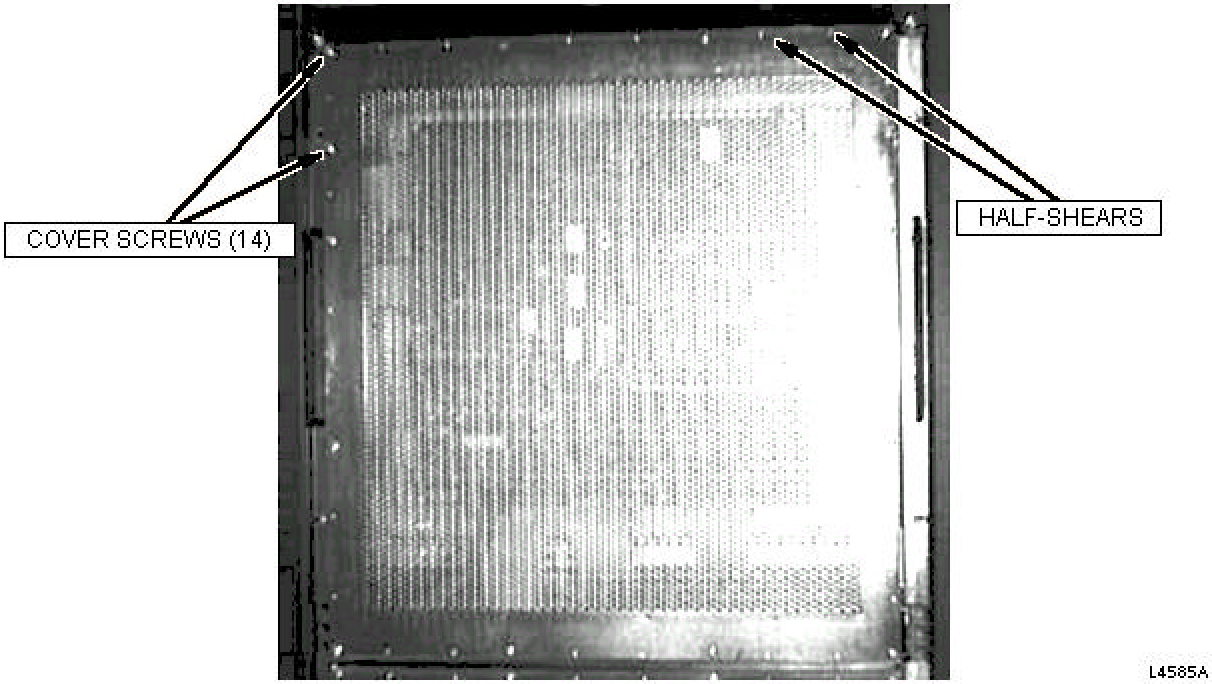
4- REPLACING THE GASM BOARD

CAUTION

Equipment damage possibility. The GRAM has static-sensitive components that may be damaged if not handled in a static-free environment. Take appropriate care (e.g., wear wrist grounding strap) when handling this module and the GASM Board.

1. Remove GRAM Cabinet front cover.

2. Use proper ESD precautions, and, at the front of the GRAM Cabinet, remove and set aside the fourteen screws that hold the electromagnetic containment (EMC) cover on the GRAM (see Illustration L4585A).

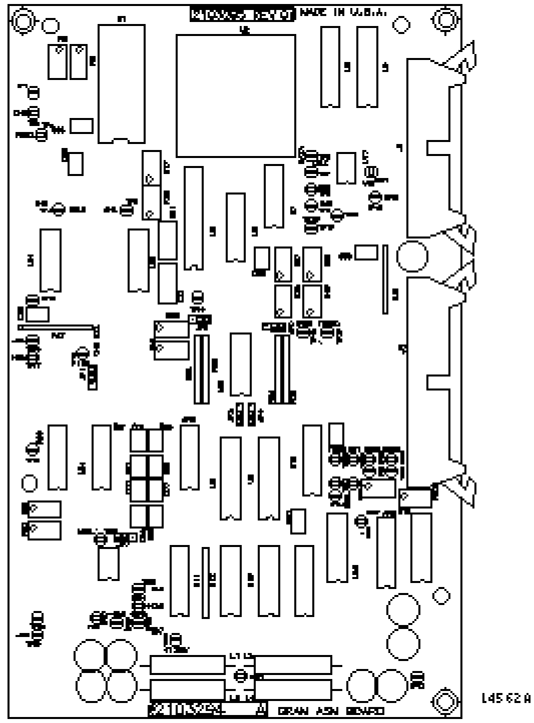


EMC COVER, SCREWS, AND HALF-SHEARS
ILLUSTRATION L4585A

Note

Importance of EMC cover screws - These fourteen screws help the EMC cover prevent electromagnetic leakage; do not misplace any of them.

3. With the EMC cover removed, disconnect the two ribbon cables on the failed GASM Board. (see Illustration L4562A).

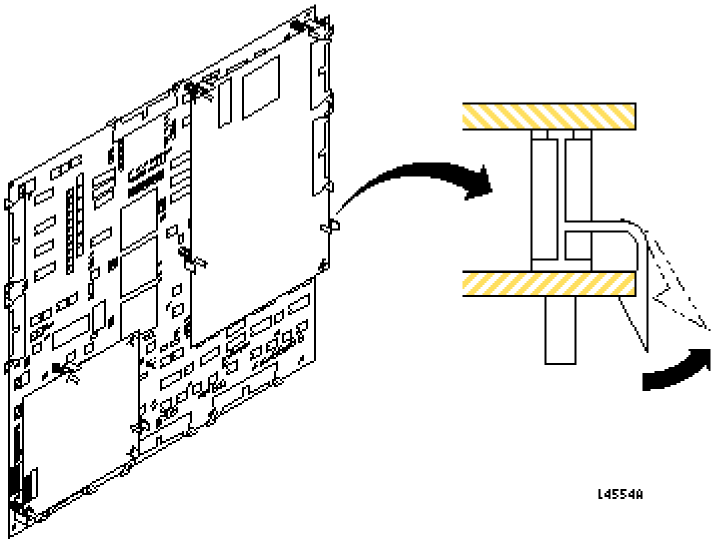


GASM CIRCUIT BOARD
ILLUSTRATION L4562A

CAUTION

Equipment damage possibility. Do not try to pull the GASM board off. The nylon stand-offs that hold it are quite brittle and may break. Push the top of each stand-off one at a time. If any of them do not push off of the board at first, work on the others interatively.

4. Carefully push back the four white nylon stand-offs on the GASM Board (see Illustration L4554A), one at a time, until all are released. Remove the board, and place it in a static bag for return to the manufacturer.



GASM BOARD AND WHITE NYLON STAND-OFFS
ILLUSTRATION L4554A

5. Remove the replacement GASM board from its static bag, and carefully attach to the GRAM. Attach the ribbon cables to the new board.

6. Reattach the EMC cover. Be sure that the half-shears (small bumps) between the screw holes face in, toward the cabinet, not out. It is important to the electromagnetic integrity of the cover that you account for all fourteen screws. Tighten them just to snug. See Illustration L4585A.

7. Replace front cover on cabinet.

8. Release the Lock Out Tag Out from the PDU for the GRAM Cabinet and the 8645 Gradient Cabinet.

5- FUNCTIONAL CHECKS REQUIRED

1. After power has been restored, check the error log for any error messages from GAP. If there are error messages relating to the GASM, follow the suggested service method in the error message.

2. Perform a head or body scan to ensure system functionality.

REVISION HISTORY

REV	DATE	AUTHOR	PRIMARY REASONS FOR CHANGE
0	May 22, 1998	J. Saperstein	Initial Conversion from Toolbook to Word